Multiclass Classification of Fetal Health Using Cardiotocogram Data

Machine Learning Mini-Project | UE24CS352A

1. Problem Statement

Cardiotocography (CTG) provides measurements describing fetal heart-rate patterns and uterine activity. This project investigates the use of supervised machine learning to classify fetal health into three categories: Normal, Suspect, and Pathological. The objective is to evaluate multiple classification algorithms and determine how effectively CTG-derived features can distinguish between these fetal-health classes.

2. Dataset and Preprocessing

The dataset initially contained 2126 instances and 21 CTG-derived input features. Data inspection found 0 missing values and 13 duplicate rows. After removing duplicate observations, 2113 instances remained. The target variable was encoded into three classes for machine learning. An 80:20 stratified train-test split was then used to preserve the class distribution across the two subsets.

Property	Value
Original samples	2126
Duplicates removed	13
Final samples	2113
Input features	21
Missing values	0

3. Exploratory Data Analysis

The dataset exhibits class imbalance. Normal cases form the majority of observations, while Suspect and Pathological cases constitute smaller portions of the dataset. This imbalance motivates the use of class-wise precision, recall, F1-score, and Macro F1-score when evaluating the classification models.

Fetal Health	Samples	Percentage
Normal	1646	77.90%
Suspect	292	13.82%
Pathological	175	8.28%

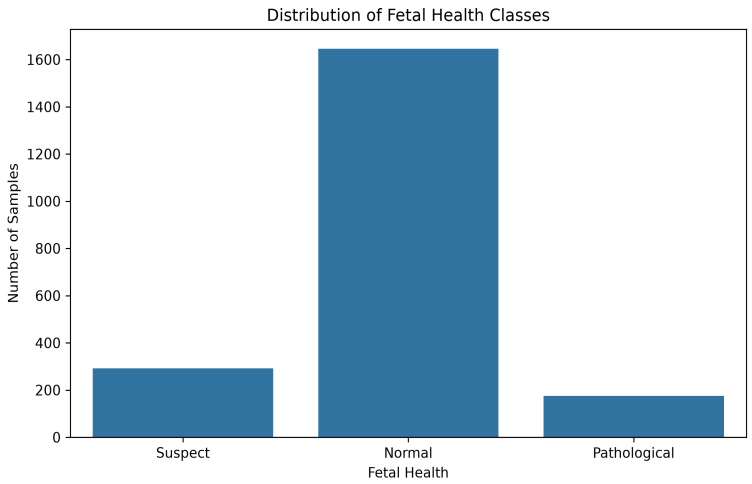


Figure 1. Distribution of fetal-health classes after preprocessing.

4. Feature Analysis

The 21 CTG features were examined using descriptive statistics, feature distributions, and a correlation matrix. No extremely strong pairwise correlations were observed during exploratory analysis. Therefore, no features were removed solely on the basis of correlation, and all 21 features were retained for the baseline classification experiments. Feature selection was reserved as a separate modeling experiment.

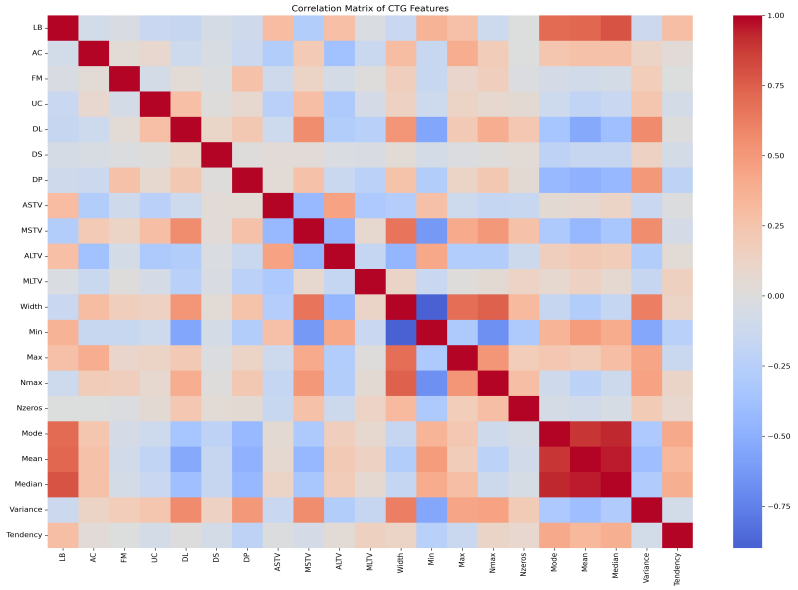


Figure 2. Correlation matrix of the 21 CTG input features.

5. Machine Learning Approach

The prepared training and testing datasets are used to evaluate multiple supervised classification algorithms. The planned models include Logistic Regression, Decision Tree, Random Forest, and XGBoost. Model performance is evaluated using accuracy, precision, recall, F1-score, Macro F1-score, and confusion matrices. The strongest-performing model is subsequently considered for hyperparameter tuning. An additional experiment compares the use of all 21 features with a reduced feature subset.

6. Model Results

Model results will be inserted after completion of the classification experiments. The final report will compare Logistic Regression, Decision Tree, Random Forest, and XGBoost using accuracy, precision, recall, F1-score, and Macro F1-score.

7. Conclusion

The preprocessing and exploratory analysis established a clean dataset containing 2,113 observations and 21 CTG input features. The analysis identified a substantial class imbalance, making class-sensitive evaluation important. The prepared dataset and stratified train-test split provide the basis for evaluating multiple machine learning approaches for multiclass fetal-health classification. Final conclusions will be based on the comparative modeling and feature-selection experiments.